6TH GRADE MATH · UNIT 4: EXPRESSIONS AND EQUATIONS

Arcade Party Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan an arcade party. They write and evaluate expressions for costs, use properties to create equivalent expressions, read exponents in prize prices, and solve one-step equations about guests and budgets.

Comparing two packages shows students why expressions with a fixed fee and a rate behave differently as the variable grows.

Standards Addressed

- 6.EE.A.1 Expressions with whole-number exponents
- 6.EE.A.2 Write, read, and evaluate expressions
- 6.EE.A.3–4 Equivalent expressions using properties
- 6.EE.B.5–7 Solve one-step equations in context

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: "Which package would you pick?" Part 1 writing and evaluating expressions.
Day 2	Part 2 properties and exponents.
Day 3	Part 3 equations and the party proposal.

Materials

- Student handout and Arcade Price Board
- Calculators
- Markers and paper for the flyer

How the Three Versions Differ

Level	What changes
CP2	Single-rate expressions, evaluating, simple exponents, and one-step equations with starters.
CP1	Multi-term expressions, package comparison tables, factoring with GCF, and budget equations with rounding.

Level	What changes
Honors	Break-even reasoning, equivalent expressions justified with properties, exponent comparisons, and designing a package to meet two conditions.

Watch For (Common Misconceptions)

- Writing 12 + g instead of 12g for "12 per guest".
- Distributing to only the first term.
- Reading 2^5 as 2 × 5.

Extension Ideas

- Graph both packages on the same coordinate grid.
- Survey classmates on their dream party and price it.
- Add sales tax of 6.25% to the final cost.

Answer Key – CP2

Part 1: Write the Costs

#	Task	Answer
1	Write an expression for the cost of game cards for g guests.	12g
2	Write an expression for the cost of game cards and drinks for g guests.	12g + 2g = 14g
3	Write an expression for the total cost of the room, game cards, and drinks.	40 + 14g
4	Evaluate your total-cost expression for 8 guests.	40 + 14(8) = 40 + 112 = $152

Part 2: Use the Properties

#	Task	Answer
5	Show that 12(g + 2) and 12g + 24 are equal when g = 5.	12(7) = 84 and 12(5) + 24 = 84
6	Pizzas cost $15. Write two equivalent expressions for the cost of p pizzas and 1 cake.	15p + 25 and 5(3p + 5)
7	A prize costs 2^5 tickets. How many tickets is that?	$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$ tickets
8	Write an expression for the number of pizzas needed for g guests if each pizza serves 4.	$\frac{g}{4}$ pizzas (round up)

Part 3: Solve and Plan

#	Task	Answer
9	The game cards cost $132. How many guests came?	12g = 132 → g = 11 guests
10	Each guest gets 50 bonus tickets. The group has 600 bonus tickets. How many guests are there?	50g = 600 → g = 12
11	Plan a party for 10 guests with game cards, drinks, the room, pizza, and cake. Find the total cost.	40 + 14(10) + 15(3) + 25 = 40 + 140 + 45 + 25 = $250 (3 pizzas feed 12)

Reflection

#	Prompt	Look for
R1	What does the variable g stand for in your expressions?	The number of guests at the party.
R2	How did you check your equation answers?	Substitute the answer back into the equation to see if both sides are equal.

Answer Key – CP1

Part 1: Package Expressions

#	Task	Answer
1	Write an expression for the total cost with the room, a game card and a drink for each guest, and one pizza for every 4 guests.	$40 + 12g + 2g + 15(\frac{g}{4}) = 40 + 14g + 3.75g = 40 + 17.75g$
2	Evaluate your expression for 8 guests and for 12 guests.	8: 40 + 142 = $182. 12: 40 + 213 = $253.
3	Pixel Palace also offers "The Mega": $20 per guest, everything included, no room fee. Write its expression and evaluate it for 8 and 12 guests.	20g; $160 and $240
4	Which package is cheaper for 8 guests? For 20 guests? What do you notice?	8: Mega ($160 < $182). 20: Mega $400 vs 40 + 355 = $395 → regular is cheaper. The regular package wins for large groups.

Part 2: Equivalent Expressions

#	Task	Answer
5	Use the distributive property to write 6(2g + 5) without parentheses. What could it describe?	12g + 30; e.g., 6 parties with 2g guest costs and $5 extra each
6	Factor 18g + 24 using the greatest common factor.	6(3g + 4)
7	Combine like terms: 12g + 40 + 2g + 25 + 3g.	17g + 65
8	A super prize costs 3^4 tickets and a mega prize costs 4^3 tickets. Which costs more and by how much?	$3^4 = 81$, $4^3 = 64$; super prize costs 17 more tickets.

Part 3: Equations & Design

#	Task	Answer
9	The Mega package cost $260. How many guests came?	20g = 260 → g = 13
10	After the room fee was paid, the family had $186 left for game cards and drinks at $14 per guest. How many guests can come?	14g = 186 → g ≈ 13.3 → 13 guests
11	Design your own package with a price per guest and a flat fee. Write its expression and an equation for a $300 budget. Solve it.	Answers vary; e.g., 50 + 15g = 300 → g = 16.67 → 16 guests (one-step after subtracting the fee is acceptable).

Reflection

#	Prompt	Look for
R1	Why do the regular and Mega packages switch which is cheaper?	The regular package has a fixed fee but a lower cost per guest; as guests increase the fixed fee matters less.
R2	How do you know two expressions are equivalent?	They give the same value for every value of the variable; properties of operations transform one into the other.

Answer Key – Honors

Part 1: Model the Packages

#	Task	Answer
1	Write the cost of the regular package (room, game card, drink, and pizza at $\frac{1}{4}$ pizza per guest) and simplify.	$40 + 12g + 2g + 15(\frac{g}{4}) = 40 + 17.75g$
2	The Mega costs 20g. For what number of guests do the two packages cost the same? (Solve by testing values or by reasoning.)	$40 + 17.75g = 20g \rightarrow 40 = 2.25g \rightarrow g \approx 17.8$. Mega is cheaper up to 17 guests; regular is cheaper from 18 guests.
3	A coupon gives $3 off each game card. Write two equivalent expressions for the new regular cost and explain each.	$40 + 9g + 2g + 3.75g = 40 + 14.75g$; or $40 + 17.75g - 3g$
4	Evaluate both of your coupon expressions for g = 16 to show they are equivalent.	$40 + 14.75(16) = 276$ and $40 + 17.75(16) - 48 = 276$

Part 2: Tickets & Powers

#	Task	Answer
5	Order these prizes by ticket cost: 2^7, 5^3, 11^2, 3^5.	$11^2 = 121$, $5^3 = 125$, $2^7 = 128$, $3^5 = 243$
6	Each guest wins an average of 2(t + 15) tickets where t is bonus tickets. Rewrite without parentheses and find the total for 12 guests with t = 50.	$2t + 30$; $2(50) + 30 = 130$ per guest × 12 = 1,560 tickets
7	Factor $48g + 36$ and $45g + 30g^2$, and explain what the factored forms show.	$12(4g + 3)$ and $15g(3 + 2g)$; the common factor is shared by every term.
8	Is $3(g + 4)$ equivalent to $3g + 4$? Use a value of g and a property to explain.	No. g = 1: $15 \neq 7$. The distributive property multiplies 3 by both terms: $3g + 12$.

Part 3: Optimize the Party

#	Task	Answer
9	How many guests can attend each package within $320? (Use equations, then round sensibly.)	Regular: $40 + 17.75g = 320 \rightarrow g \approx 15.8 \rightarrow$ 15 guests. Mega: $20g = 320 \rightarrow$ 16 guests.
10	With a $3-off coupon, recompute for the regular package. Which package now allows more guests?	$40 + 14.75g = 320 \rightarrow g \approx 18.98 \rightarrow$ 18 guests; the coupon regular package allows more guests.
11	Design a package for Pixel Palace that is cheaper than the Mega for 10 guests but costs more than the regular package for 25 guests. Write its expression and show both comparisons.	Answers vary. Check: cost for 10 guests < $200 (Mega) and cost for 25 guests > $483.75 (regular). Example: 19.5g gives $195 and $487.50.

Reflection

#	Prompt	Look for
R1	Explain the break-even point in words a parent would understand.	It is the number of guests where both packages cost the same; below it one is cheaper, above it the other.
R2	Why must you round down guest counts when solving budget equations?	Rounding up would cost more than the budget; only whole guests that fit the budget are allowed.